by multiplying by $\sqrt{t^*}$. Recall that t^* is expressed in volume time units where $t^* = 1$ represents a one-day time period (volume-time). And since the order size is decreasing in each period, timing risk needs to be further adjusted downward by the $\sqrt{1/3}$ factor (see derivation below). This value is converted to basis points by multiplying by $10^4 bp$.

Therefore, timing risk is expressed in terms of POV and α following Equations 7.8 and 7.9 respectively. This is:

$$TR_{bp}(POV) = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \frac{1 - POV}{POV}} \cdot 10^4 bp \tag{7.21}$$

$$TR_{bp}(\alpha) = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \alpha^{-1}} \cdot 10^4 bp \tag{7.22}$$

These values expressed in terms of dollars follow directly from above:

$$TR_{\$}(POV) = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \frac{1 - POV}{POV}} \cdot X \cdot P_0 \tag{7.23}$$

$$TR_{\$}(\alpha) = \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \alpha^{-1}} \cdot X \cdot P_0 \tag{7.24}$$

The reason the timing risk equations simplify so nicely is that the POV and α strategies assume a constant trading rate. Timing risk for a trade schedule, however, is not as nice. It is slightly more complicated since we need to estimate the risk for each period. This is as follows:

Let,

$r_k =$ number of unexecuted shares at the beginning of period k
$r_k = \sum_{j=k}^{n} x_j$
$d =$ number of trading periods per day
$v_k =$ expected volume in period k excluding the order shares
$\left(\sigma^2 \cdot \frac{1}{250} \cdot \frac{1}{d}\right) =$ price variance scaled for the length of a trading period
$P_0 =$ stock price at the beginning of the trading period

Timing risk for a trade schedule is the sum of the dollar risk in each trading period. That is:

$$TR_{\$}(x_k) = \sqrt{\sum_{k=1}^{n} r_k^2 \cdot \sigma^2 \cdot \frac{1}{250} \cdot \frac{1}{d} \cdot P_0^2} \tag{7.25}$$

In this notation, σ^2 is expressed in $(\$/\text{share})^2$ units and scaled for the length of the trading period. We divide by 250 to arrive at the volatility for a day and then further divide by the number of periods per day d. For example, if we break the day into 10 equal periods of volume we have $d = 10$. Finally, multiplying by P_0^2 converts volatility from $(\text{return})^2$ units